

Learning and Behavior: Classical Conditioning

Learning and Behavior (PSY/NSC 3320)

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Classical Conditioning

“A type of learning in which an initially neutral stimulus—the conditioned stimulus (CS)—when paired with a stimulus that elicits a reflex response—the unconditioned stimulus (US)—results in a learned, or conditioned, response (CR) when the CS is presented” (*APA*)

AKA Pavlovian Conditioning

Classical Conditioning

Pairing a reflex stimulus (US) with a neutral stimulus (CS) to form a learned response (CR)

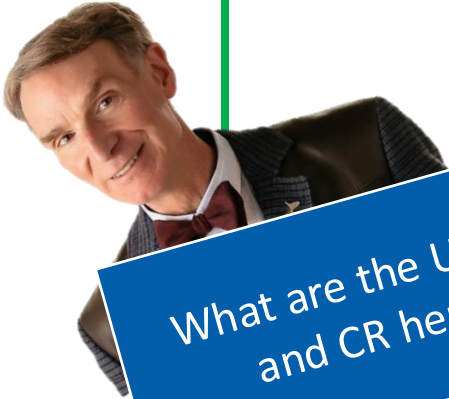
Key Terms

Unconditioned Stimulus (US) – reflex stimulus

Conditioned Stimulus (CS or NS) – neutral stimulus

Conditioned Response (CR) – learned association between CS and reflex

CS : US → R



What are the US, CS,
and CR here?

Key Terms

Unconditioned Stimulus (US) – reflex stimulus

Conditioned Stimulus (CS or NS) – neutral stimulus

Conditioned Response (CR) – learned association between CS and reflex

- | | | | |
|----|------------------|---|------------|
| 1. | Food | → | |
| 2. | Metronome | X | |
| 3. | Metronome : Food | → | Salivation |
| 4. | Metronome | → | |



Is this example one of appetitive or aversive conditioning?

Key Terms

Appetitive Conditioning – US is appetitive stimulus
Aversive Conditioning – US is aversive stimulus

- | | | | |
|----|------------------|---|------------|
| 1. | Food | → | |
| 2. | Metronome | X | |
| 3. | Metronome : Food | → | Salivation |
| 4. | Metronome | → | |

Key Terms

Excitatory Conditioning – CS predicts US
Inhibitory Conditioning – CS predicts *absence* of US



Unconditioned Response
(Salivation)



Unconditioned Stimulus
(Food)



No Response



Neutral Stimulus
(Bell Ringing)



Unconditioned Response
(Salivation)



Neutral Stimulus
(Bell Ringing)



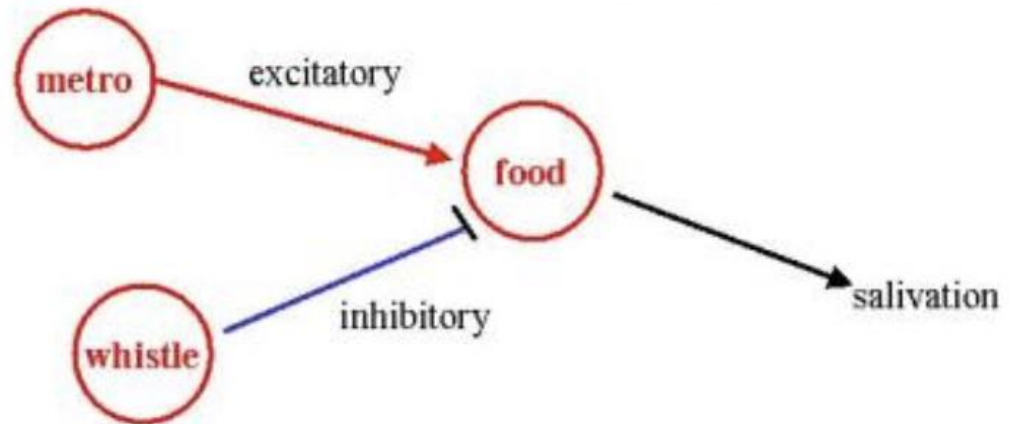
Unconditioned Stimulus
(Food)



Conditioned Response
(Salivation)



Conditioned Stimulus
(Bell Ringing)



Key Terms

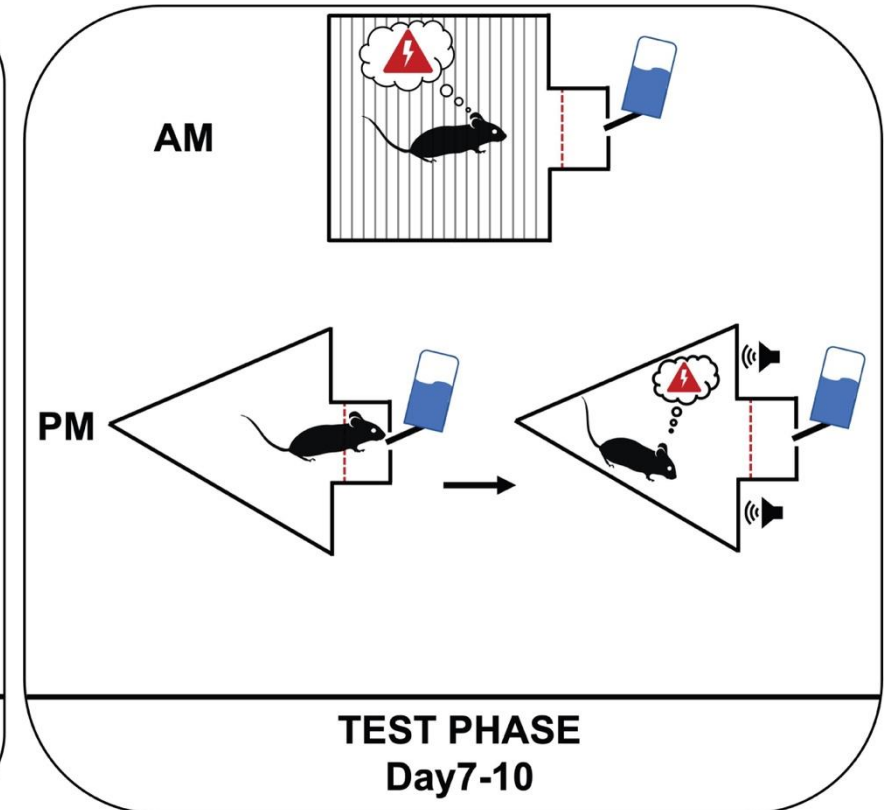
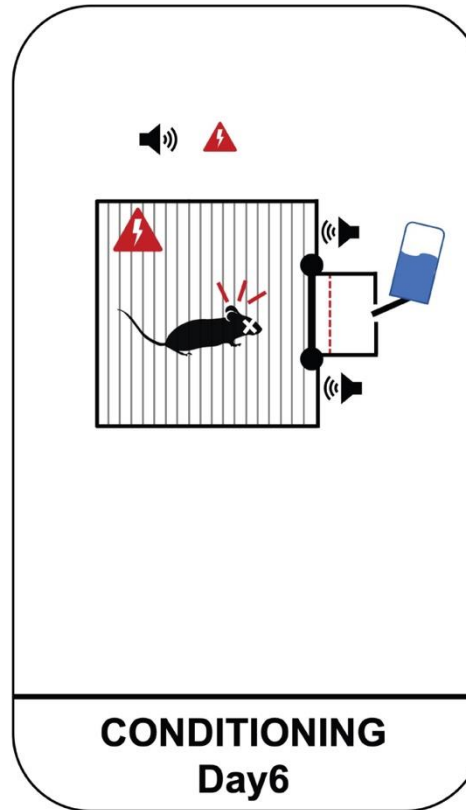
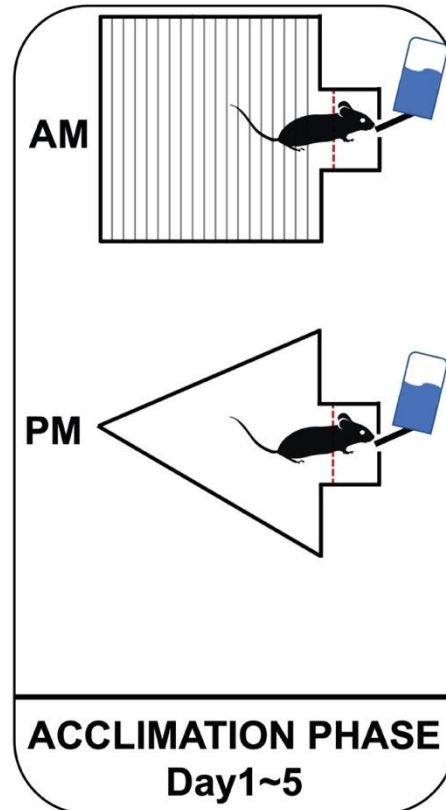
Excitatory Conditioning – CS predicts US

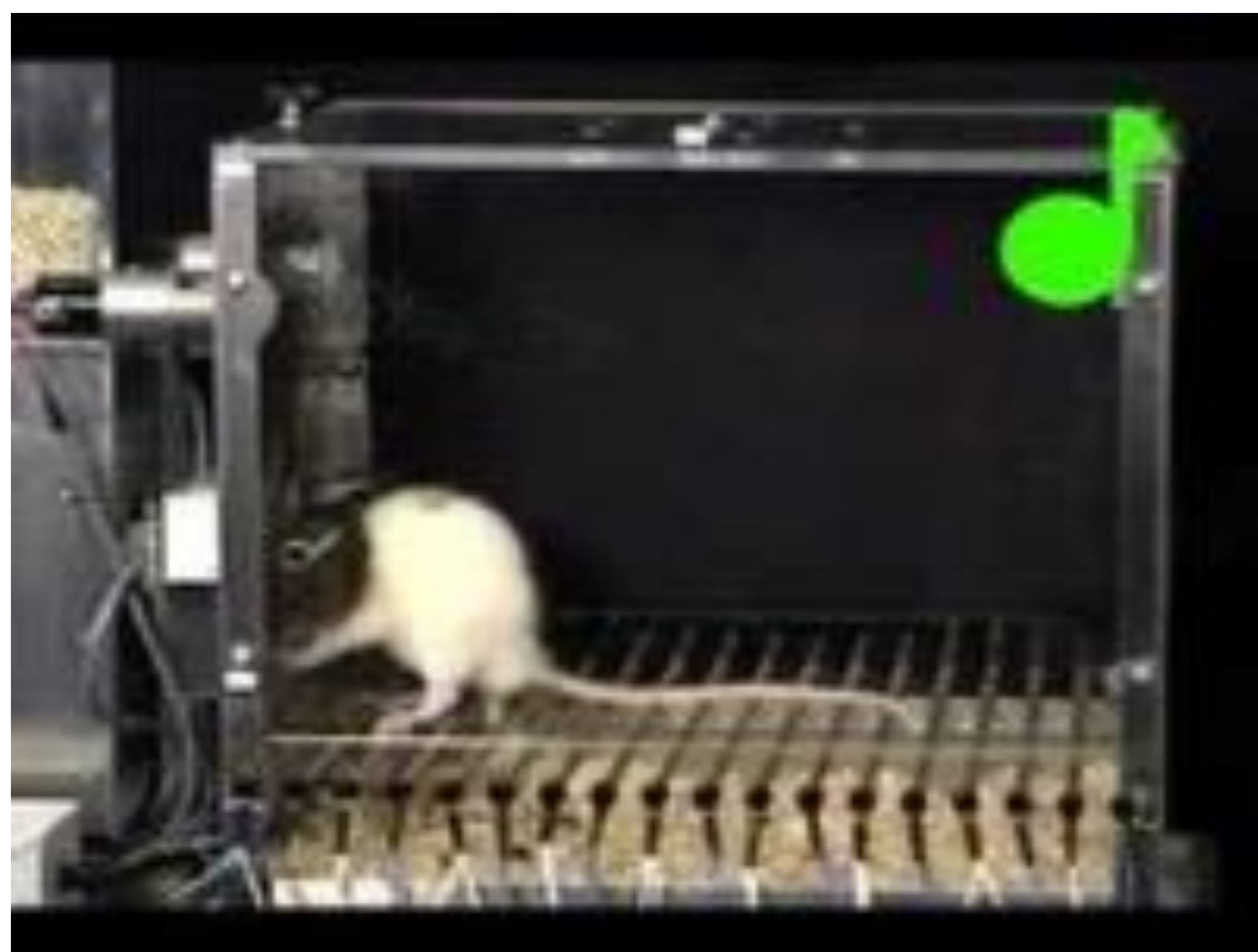
Inhibitory Conditioning – CS predicts *absence* of US

Conditioned Suppression

A paradigm that reduces an ongoing behavior after a conditioned stimulus

Conditioned
Suppression
≠
Inhibitory
Conditioning





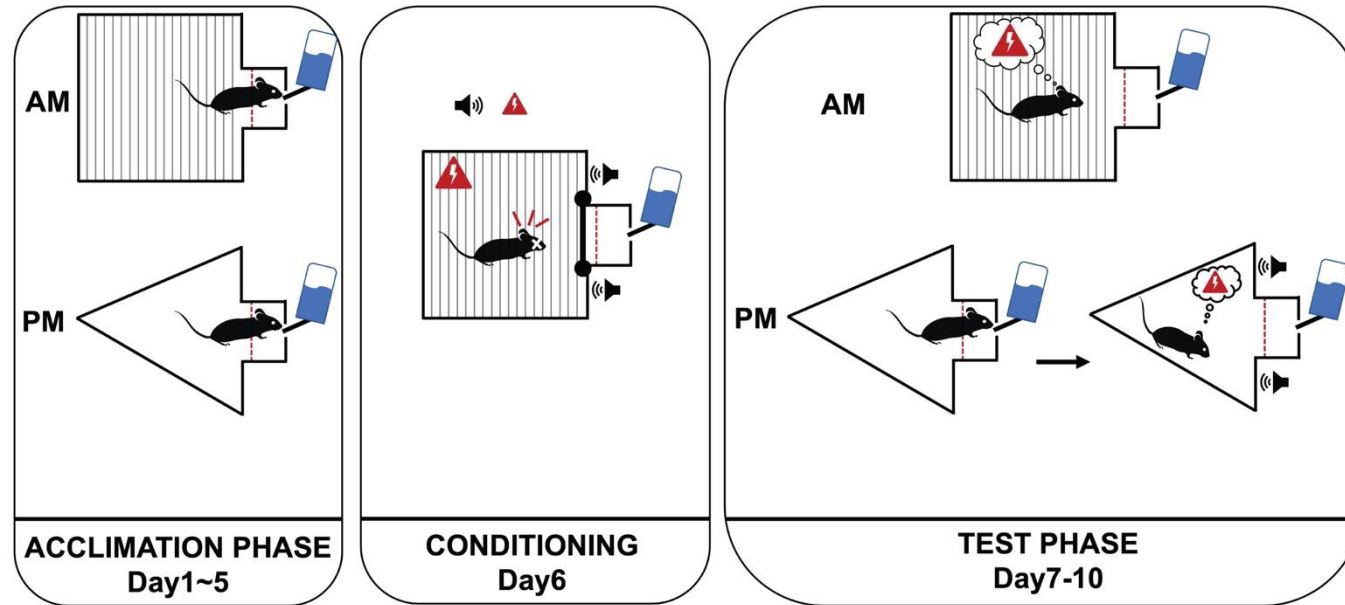
Key Terms

Excitatory Conditioning – CS predicts appetitive US

Inhibitory Conditioning – CS predicts *absence* of appetitive US

Conditioned Suppression

A paradigm that reduces an ongoing behavior after a conditioned stimulus



$$\text{Suppression Ratio} = \frac{5}{5 + 20} = 0.2$$

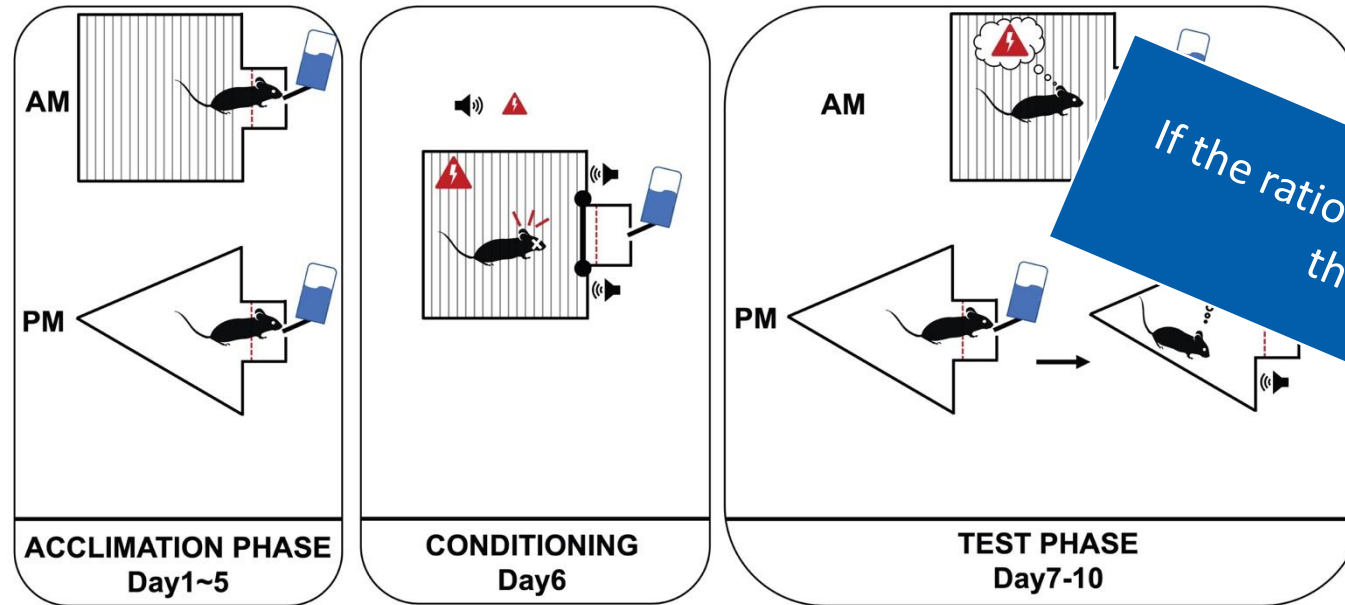
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Conditioned Suppression

A paradigm that reduces an ongoing behavior after a conditioned stimulus

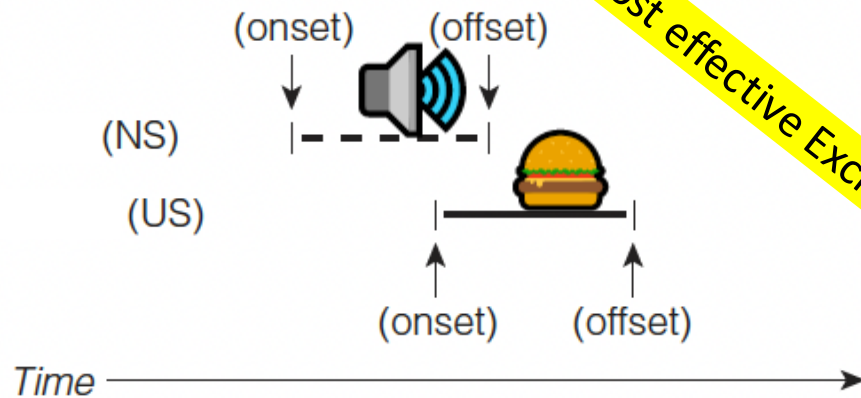


If the ratio were 0, what would that mean?

$$\text{Suppression Ratio} = \frac{5}{5 + 20} = 0.2$$

Classical Conditioning Schedules

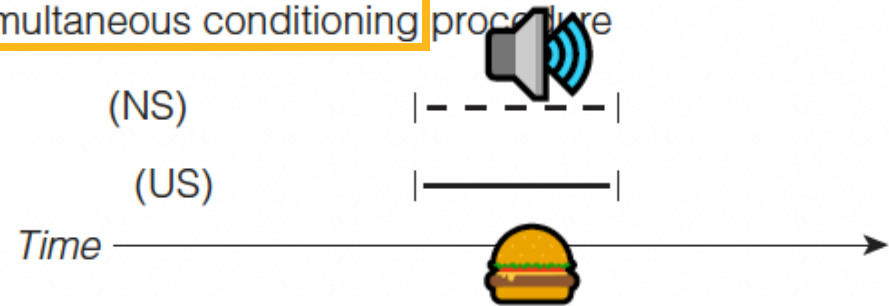
(a) Delayed conditioning procedure



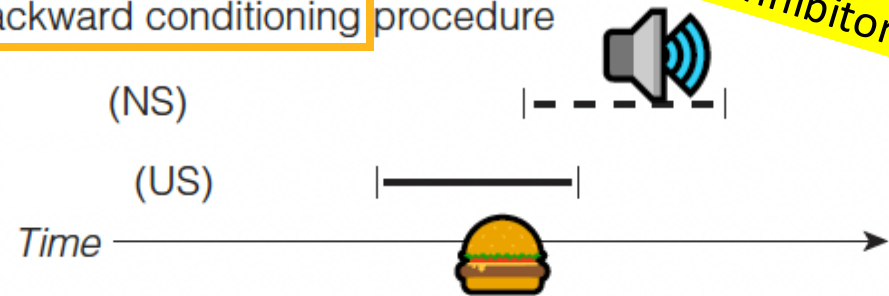
(b) Trace conditioning procedure



(c) Simultaneous conditioning procedure

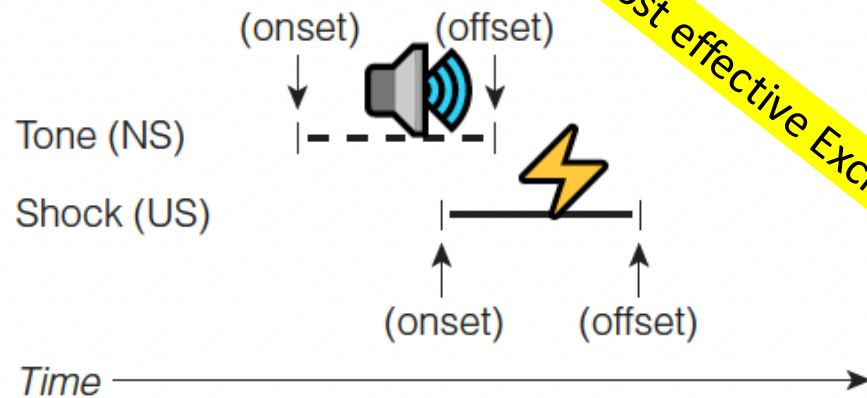


(d) Backward conditioning procedure



Classical Conditioning Schedules

(a) Delayed conditioning procedure

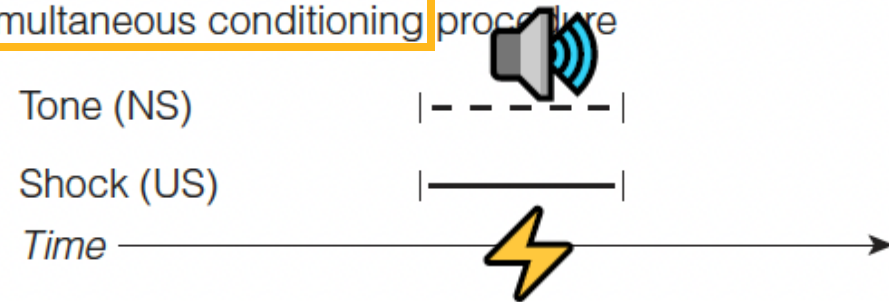


Most effective Excitatory

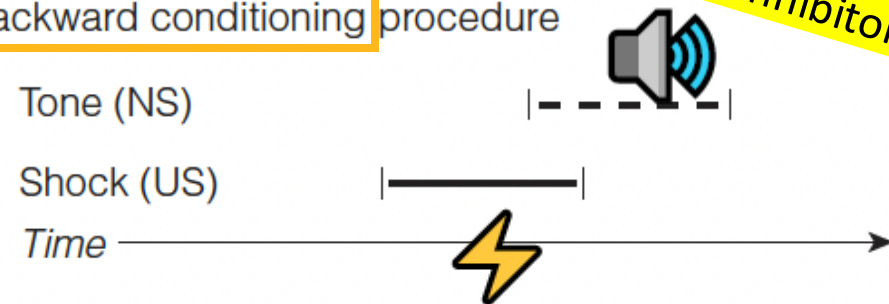
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(c) Simultaneous conditioning procedure



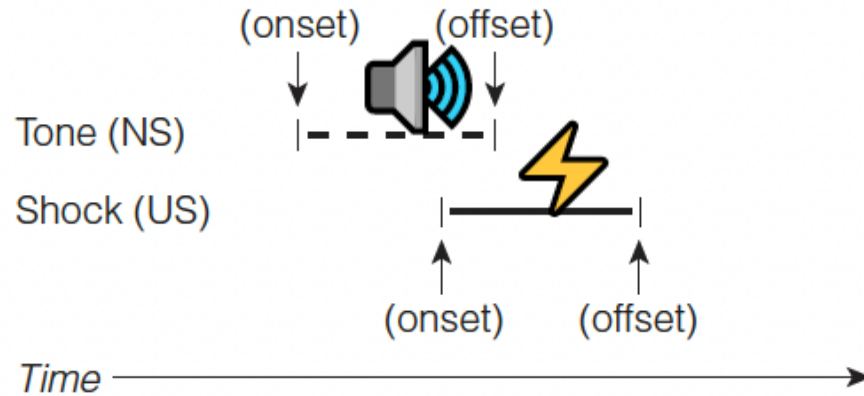
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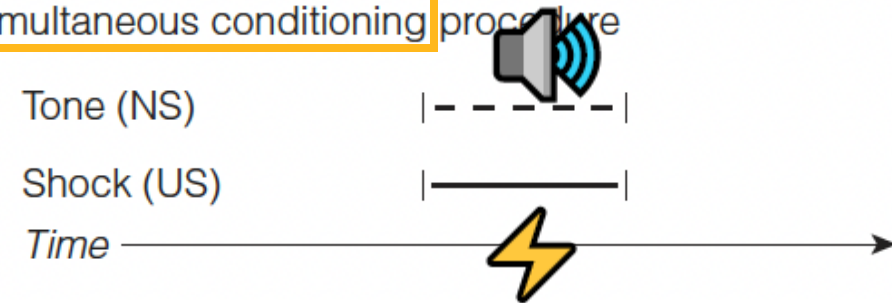
A form of Inhibitory

Classical Conditioning Schedules

(a) Delayed conditioning procedure



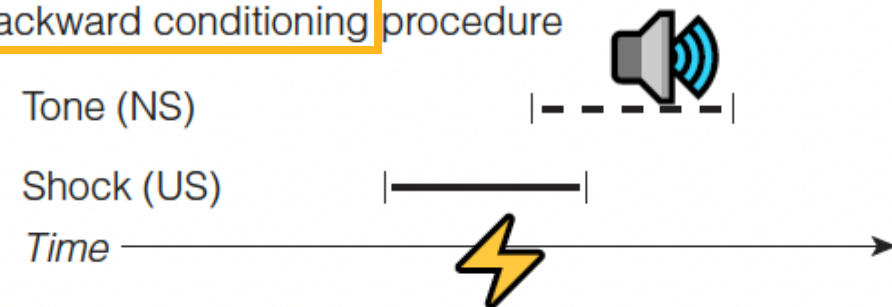
(c) Simultaneous conditioning procedure



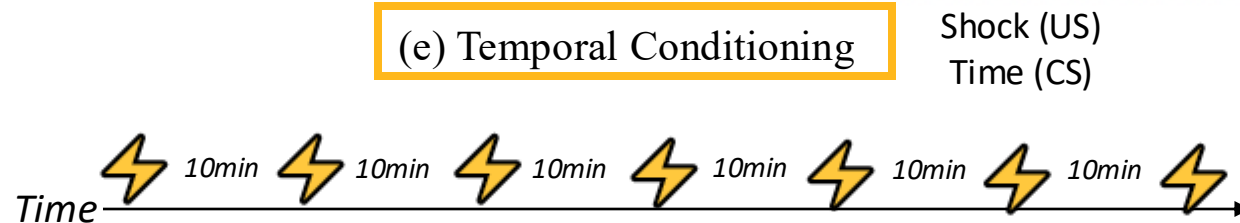
(b) Trace conditioning procedure



(d) Backward conditioning procedure



(e) Temporal Conditioning



Measurements of Interest

Acquisition

Extinction

Generalization

Discrimination

Measurements of Interest

Acquisition – process of creating/strengthening learned CS-US relationship

Extinction – process of destroying/weakening learned CS-US relationship

Generalization – tendency for stimulus *similar* to CS to elicit R

Discrimination – tendency for one CS alone, and no stimuli like it, to elicit R

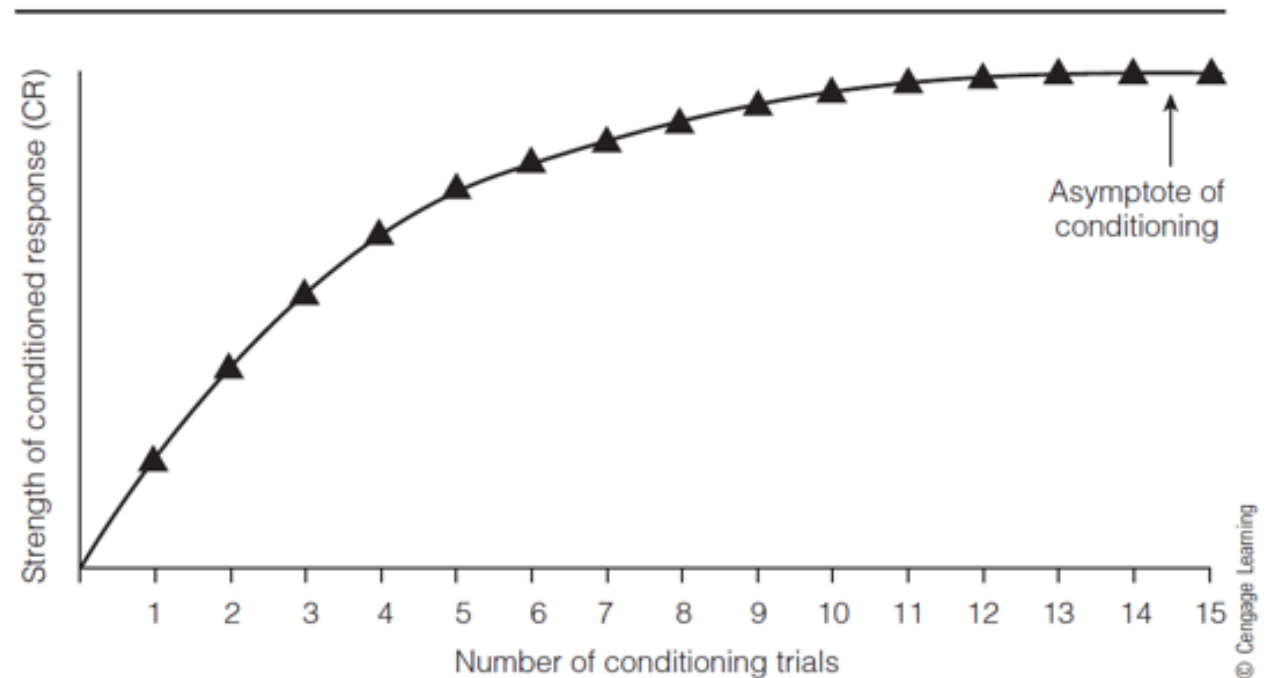
Measurements of Interest

Acquisition – process of creating/strengthening learned CS-US relationship

Asymptote of conditioning – maximum R to CS (max. conditioning)

Stronger US/CS → stronger & faster acq.

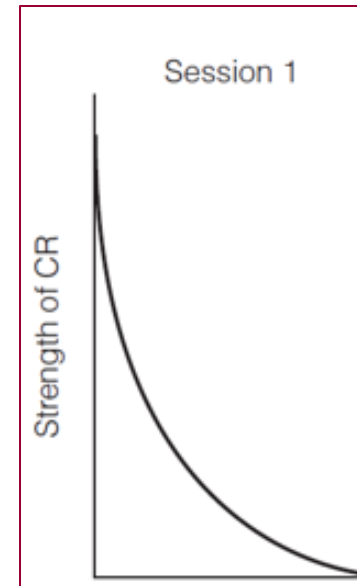
- Larger amount of food or preferred food (meat vs kibble)
- Louder vs softer tone



Measurements of Interest

Extinction – process of destroying/weakening learned CS-US relationship

Stop pairing CS/US
(present CS alone)
→
relationship extinguishes



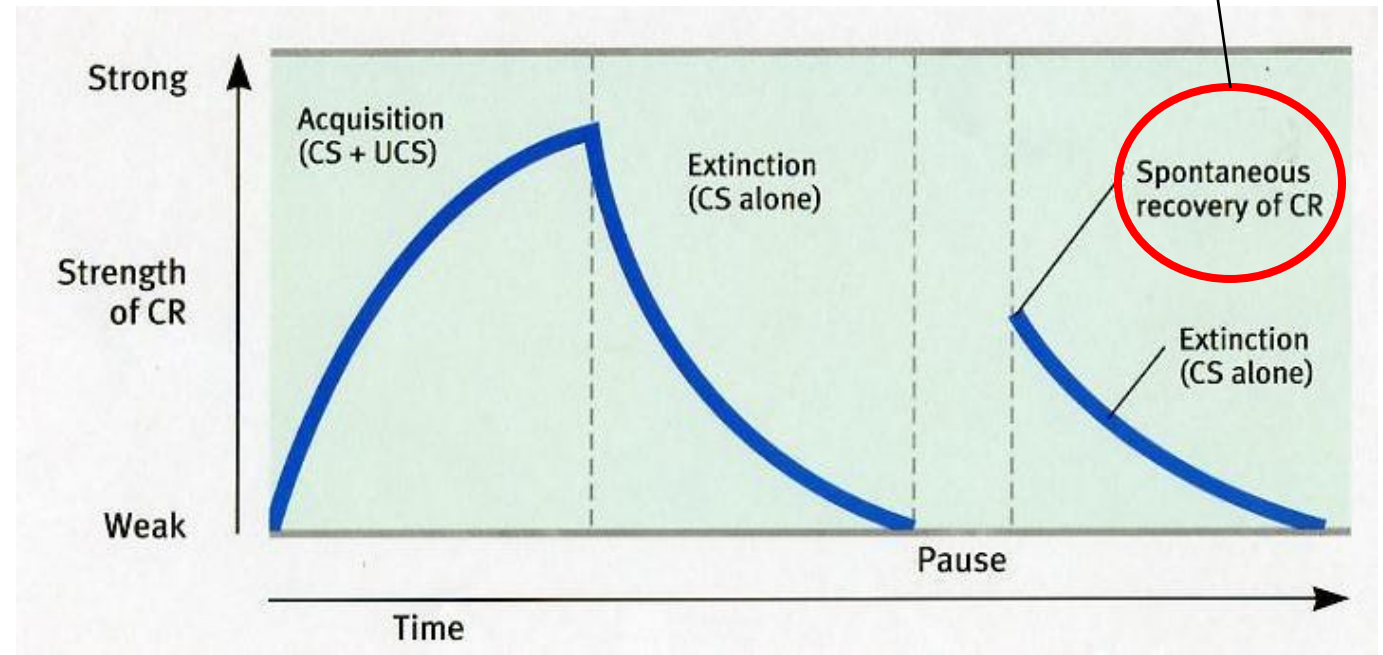
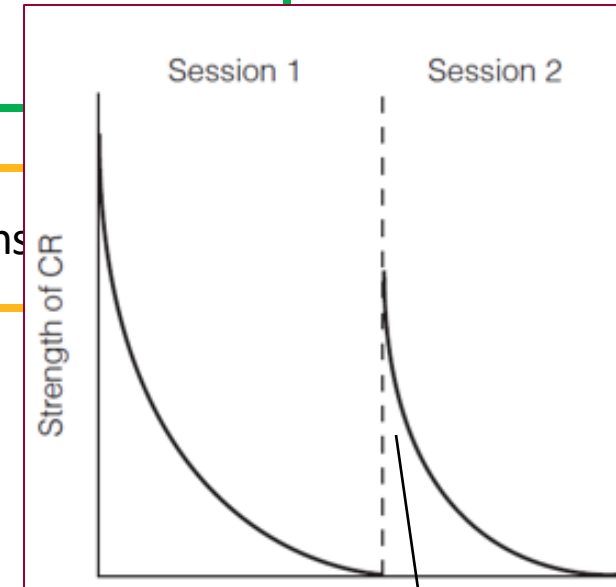
Measurements of Interest

Extinction – process of destroying/weakening learned CS-US relations

Stop CS/US pairings
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Spontaneous Recovery

After conditioning & extinction, pair CS/US again
and subject spontaneously recovers relationship



Measurements of Interest

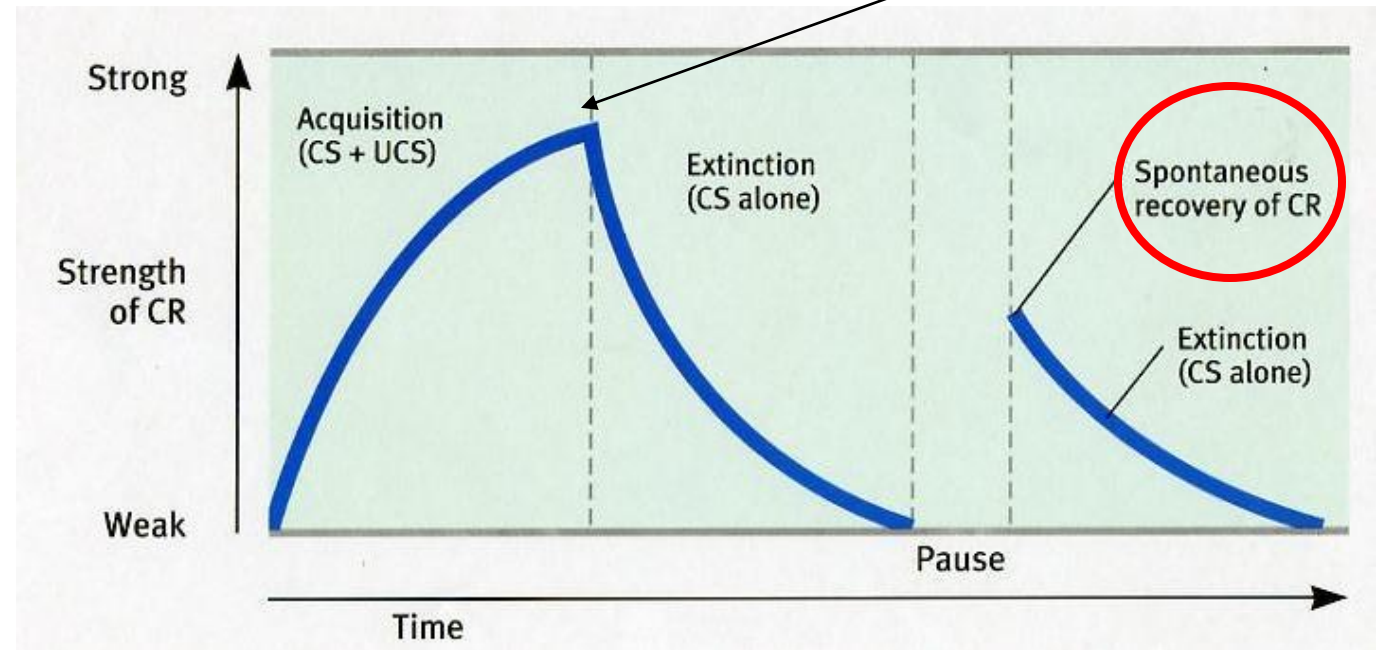
Extinction – process of destroying/weakening learned CS-US relationships

Acquisition is flattening out at the top of the curve... What is it approaching?

Stop CS/US pairings
(present CS alone)
→
relationship extinguishes

Spontaneous Recovery

After conditioning & extinction, pair CS/US again
and subject spontaneously recovers relationship



Measurements of Interest

Generalization – tendency for stimulus *similar* to CS to elicit R
More similar stimulus, stronger response...

Little Albert



WHITE RAT
NS



NO RESPONSE



LOUD NOISE
UCS



FEAR RESPONSE
UCR



LOUD NOISE + WHITE RAT
UCS + NS



FEAR RESPONSE
UCR



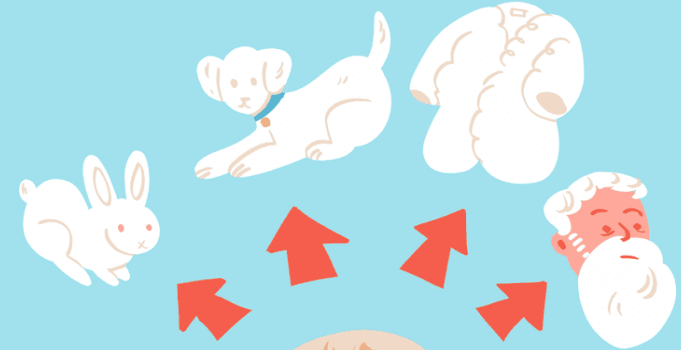
WHITE RAT
CS



FEAR RESPONSE
CR

By Louise Graham
1652

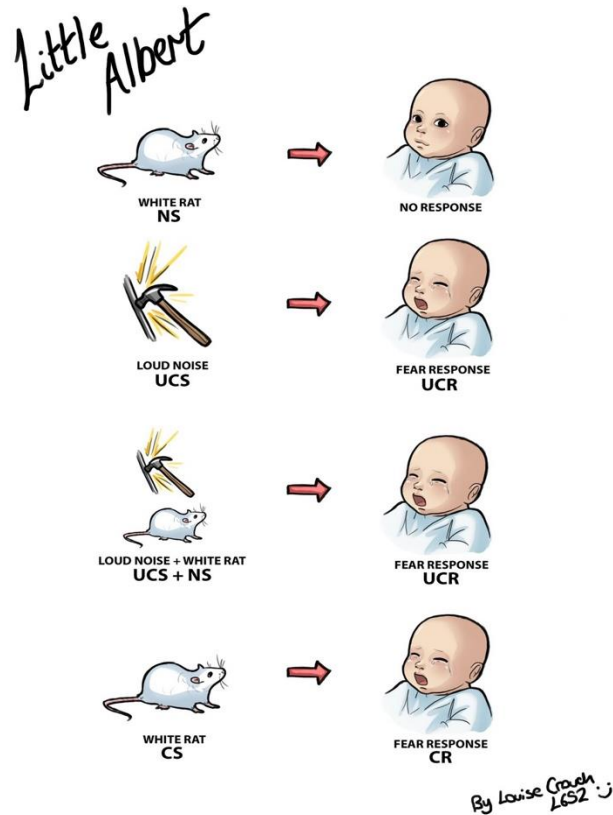
**A conditioned response to one object
is exhibited in the presence of similar stimuli**



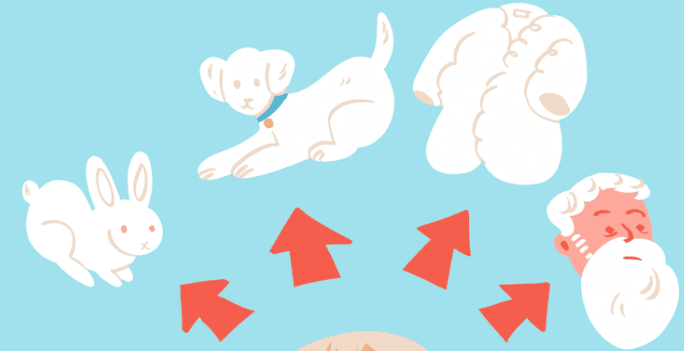
Measurements of Interest

Generalization – tendency for stimulus *similar* to CS to elicit R
More similar, stronger response...

Semantic Generalization – Generalization can be through *meaning/concept*, too:
“Car” : Shock → Fear; “Car” → Fear; “Truck” → Fear



A conditioned response to one object is exhibited in the presence of similar stimuli



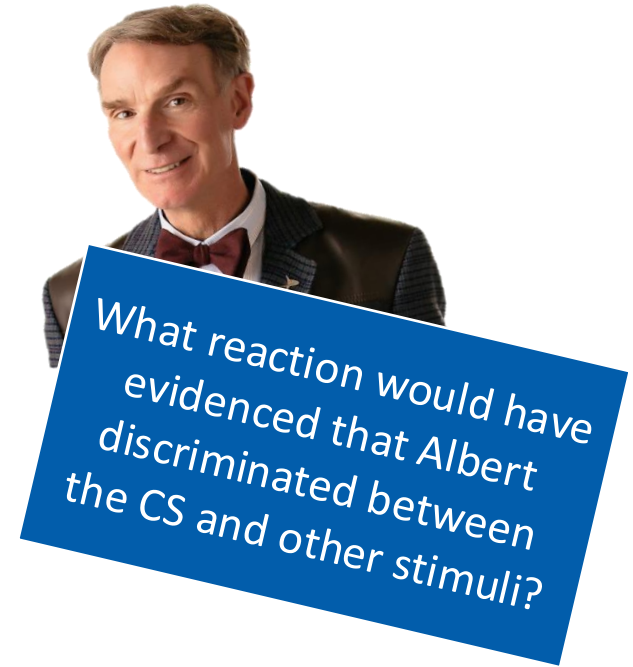
Measurements of Interest

Generalization – tendency for stimulus *similar* to CS to elicit R



Measurements of Interest

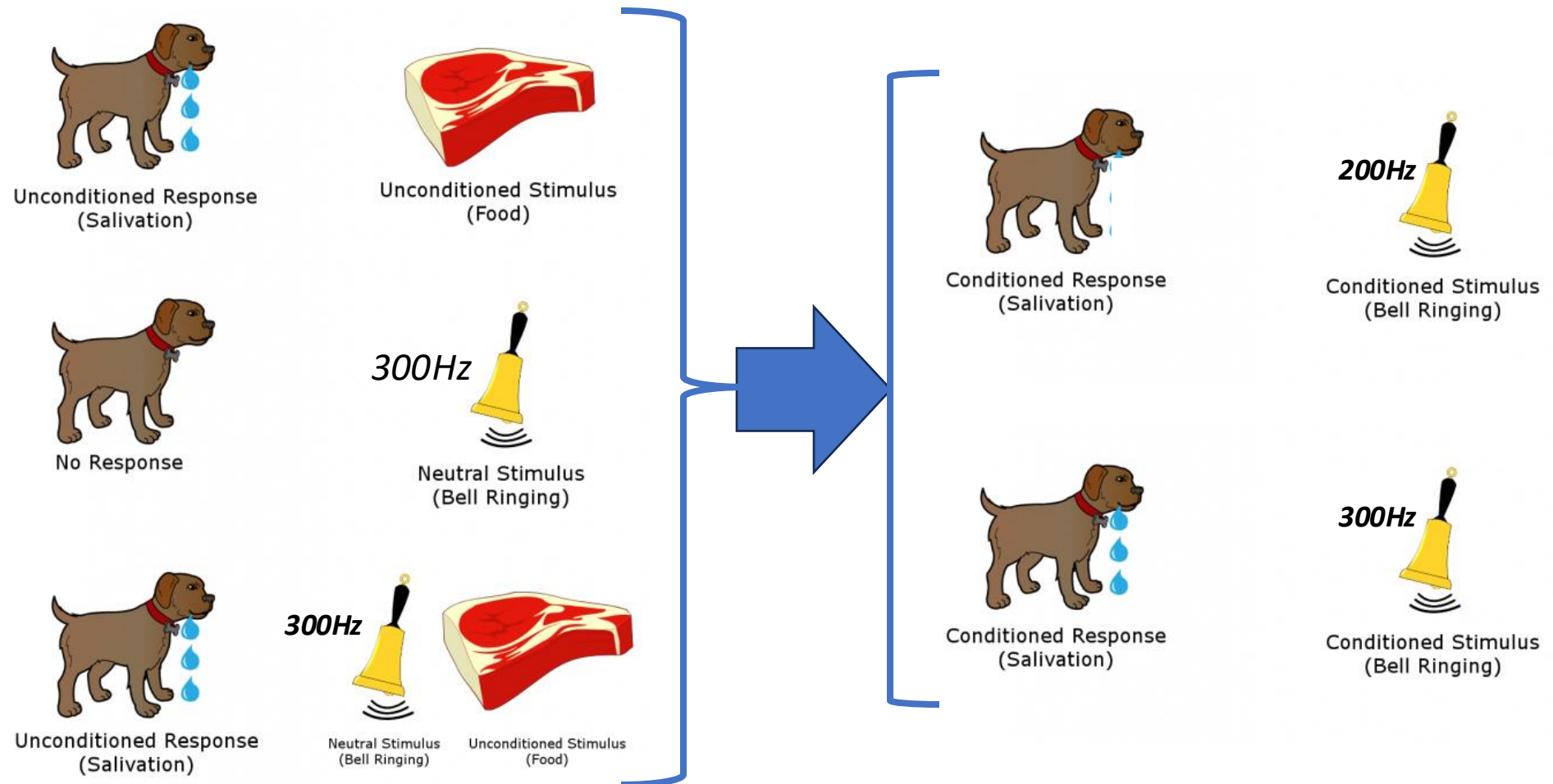
Discrimination – tendency for one CS alone, and no stimuli like it, to elicit R



What reaction would have evidenced that Albert discriminated between the CS and other stimuli?

Measurements of Interest

Discrimination – tendency for one CS alone, and no stimuli like it, to elicit R



Considerations in Classical Conditioning

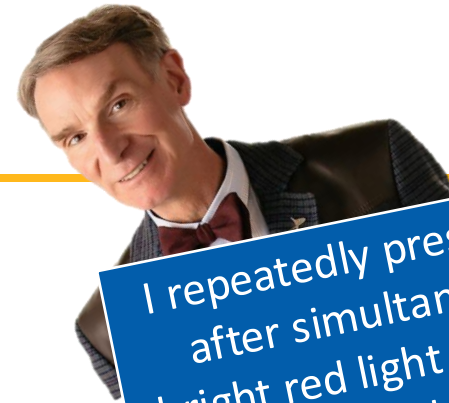
Overshadowing – when simultaneously presented, the greater of two neutral stimuli will tie to a R

Blocking – when conditioning a new CS-US, presenting a previous CS of that US will impair the process

Latent Inhibition – unknown stimuli make better CS than known stimuli

Considerations in Classical Conditioning

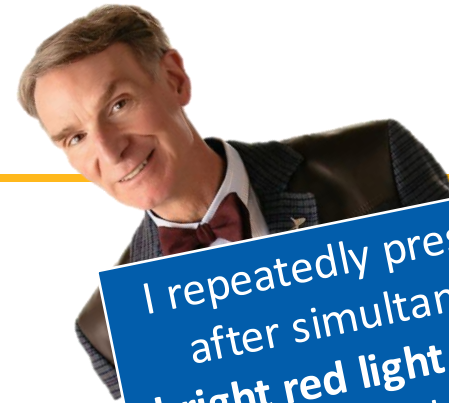
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I repeatedly present a dog with food after simultaneously displaying a bright red light and a dim green light. Which stimulus will elicit salivation after conditioning?

Considerations in Classical Conditioning

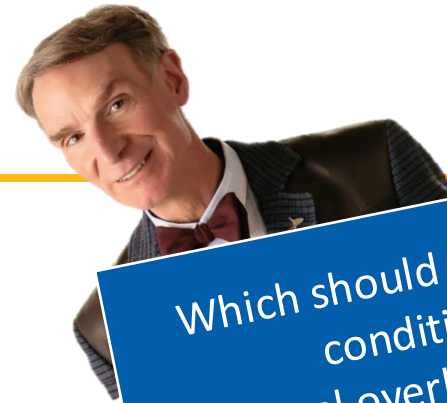
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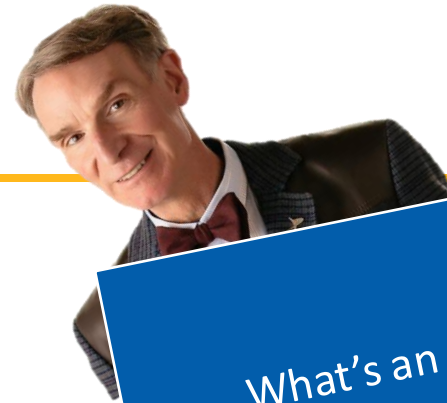
- Which should I use as a CS in my new conditioning paradigm:
- normal overhead light in my rat's cage
 - a red light the rat hasn't seen before

Considerations in Classical Conditioning

Overshadowing – when simultaneously presented, the greater of two neutral stimuli will tie to a R

Blocking – when conditioning a new CS-US, presenting a previous CS of that US will impair the process

Latent Inhibition – unknown stimuli make better CS than known stimuli



What's an example of blocking?

Considerations in Classical Conditioning

Overshadowing – when simultaneously presented, the greater of two stimuli will tie to the R

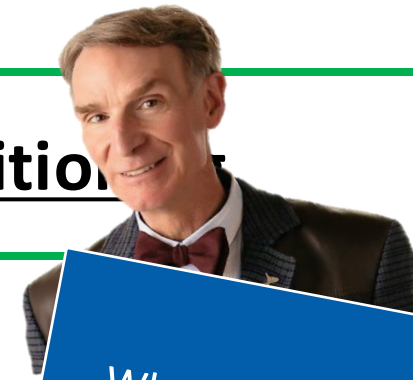
Blocking – when conditioning a new CS-US, presenting a previous CS of that US will impair the process

Latent Inhibition – unknown stimuli make better CS than known stimuli

Context Matters:

If conditioning always occurs with a certain contextual stimulus, that stimulus will be necessary to elicit the CR – in this case, that contextual stimulus is an **occasion setter**

Considerations in Classical Conditioning



What's an example of an occasion setter?

Overshadowing – when simultaneously presented, the greater of two stimuli

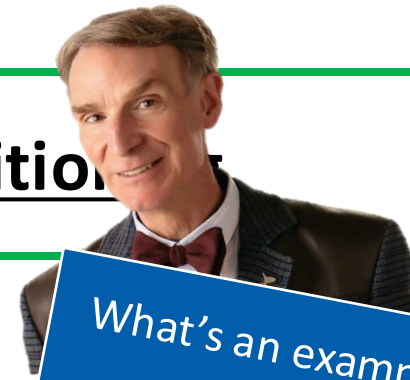
Blocking – when conditioning a new CS-US, presenting a previous CS of that US will impair the process

Latent Inhibition – unknown stimuli make better CS than known stimuli

Context Matters:

Contextual stimuli can be an **occasion setter** for learned response to CS

Considerations in Classical Conditioning



What's an example of an occasion setter?

Classroom {Tres' Voice : bored → sleepy

Classroom {Tres' Voice → sleepy

No Classroom {Tres' Voice X sleepy

Overshadowing – when simultaneously presented, the greater of 1

Blocking – when conditioning a new CS-US, presenting a previous CS of

Latent Inhibition – unknown stimuli make better CS than known stimuli

Context Matters:

Contextual stimuli can be an **occasion setter** for learned response to CS

Considerations in Classical Conditioning



Could be more specific than a setting, even just a certain person or noise being present could be an **occasion setter**

Overshadowing – when simultaneously presented, the greater of the two CSs dominates the response

Blocking – when conditioning a new CS-US, presenting a previous CS or context that predicts the US

Latent Inhibition – unknown stimuli make better CS than known stimuli

Context Matters:

Contextual stimuli can be an **occasion setter** for learned response to CS

Questions?